

Food Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen

**Total marks:
100**

Section I – 20 marks (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–18)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Mechanisation is used by sectors of the food manufacturing industry because it

 - A. meets consumer demands.
 - B. requires minimal maintenance.
 - C. provides employment opportunities.
 - D. assists in maintaining product consistency.

- 2** A company that delivers hot, ready-to-eat family meals belongs to which sector of the Australian food industry?

 - A. Food retail
 - B. Agriculture and fisheries
 - C. Food service and catering
 - D. Food processing/manufacturing

- 3** Which is the best nutritional advice for a teenager who wants to avoid developing osteoporosis in the future?

 - A. Eat dairy foods
 - B. Spend time in the sun
 - C. Take a calcium supplement
 - D. Increase consumption of red meat

- 4** At which stage of the food product development process would storage trials be conducted?

 - A. Technical feasibility
 - B. Product specifications
 - C. Testing product prototype
 - D. Development of a prototype

- 5 What is the role of anti-caking agents in packaged foods?
- A. To allow the product to flow freely
 - B. To prevent food from drying out
 - C. To improve the nutritive value
 - D. To enhance the appearance
- 6 The best time to conduct a SWOT analysis is when
- A. a design brief is written.
 - B. consumer needs are identified.
 - C. resources and anticipated returns are analysed.
 - D. the manufacturer analyses the opportunities in the marketplace.
- 7 A food processing company is experimenting with organic green bananas to produce gluten-free flour.
- What type of food product development is this?
- A. Me too
 - B. Copycat
 - C. New to world
 - D. Line extension
- 8 The agriculture and fisheries sector launches a smart phone application for consumers to identify ethically-sourced raw materials used in food production.
- This is an example of
- A. quality assurance.
 - B. research and development.
 - C. idea generation and screening.
 - D. emerging technologies in food production.

- 9** Due to a disease outbreak, raw prawn imports into Australia have been suspended.

Who has the responsibility for enforcing this legislation?

- A. SafeWork NSW
- B. NSW Office of Water
- C. Department of Immigration
- D. Department of Agriculture and Water Resources

- 10** A packaged frozen chicken is left on the kitchen sink at the start of the day to defrost in time for the evening meal.

Which of the following is most likely to cause the chicken to become unsafe for consumption?

- A. Light
- B. Infestation
- C. Temperature
- D. Enzyme activity

- 11** A food manufacturer launches a new product into the market. Initially they set the price quite high.

This is an example of

- A. price skimming.
- B. penetration pricing.
- C. promotional pricing.
- D. discriminatory pricing.

- 12** Which process do manufacturers use to ensure that raw materials are standardised?

- A. HACCP
- B. Quality control
- C. Sensory testing
- D. Quality assurance

- 13** Which dish could a vegan eat to assist in preventing anaemia?
- A. Macaroni and cheese
 - B. Grilled fish and salad
 - C. Fruit salad with coconut milk yoghurt
 - D. Chickpea, spinach and citrus salad
- 14** Why is fibre classified as an active non-nutrient?
- A. It is essential in the diet.
 - B. It provides fuel for energy.
 - C. It is not absorbed in the digestive system.
 - D. It provides a range of vitamins and minerals.
- 15** Which of the following is an example of a value-added product in the agriculture and fisheries sector?
- A. Barbequed chicken
 - B. Frozen cooked prawns
 - C. Kids' snack pack of apples
 - D. Cheese and bacon bread rolls
- 16** Nut-free policies are becoming increasingly popular in institutions where food is consumed.

What is the reason for this?

- A. Nutritional concerns
- B. The risk of anaphylaxis
- C. Workplace health and safety
- D. High levels of food intolerance

- 17** Which piece of equipment would be suitable for the commercial manufacture of tomato sauce?
- A. Retort
 - B. Atomiser
 - C. Centrifuge
 - D. Tunnel oven
- 18** Which piece of legislation regulates the labelling of free range eggs?
- A. *Trade Practices Act 1974* (Cth)
 - B. *Competition and Consumer Act 2010* (Cth)
 - C. *Food Standards Australia New Zealand Act 1991* (Cth)
 - D. *Protection of the Environment Operations Act 1997* (NSW)
- 19** Which process uses the principles of fermentation?
- A. Drying
 - B. Pickling
 - C. Jam making
 - D. Sourdough bread making
- 20** Which of the following contributes to the risk of cardiovascular disease?
- A. Omega 3 fatty acids
 - B. Low density lipoproteins (LDL) cholesterol
 - C. High density lipoproteins (HDL) cholesterol
 - D. A diet rich in sugar and monounsaturated fatty acids

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HIGHER SCHOOL CERTIFICATE
EXAMINATION

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Centre Number

Food Technology

Sections II, III and IV Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page
- Use this answer booklet for Sections II, III and IV
- Follow the instructions inside this booklet as to where you should write your answers

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (9 marks)

- (a) Complete the following table.

2

<i>Sector of the Australian food industry</i>	<i>Food that reflects the sector</i>
Agriculture and fisheries	
	Pre-packaged salad mix
Food processing/manufacturing	

- (b) Outline the role of advisory groups within the Australian food industry. Use a specific example to support your answer.

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Question 21 continues on page 11

Question 21 (continued)

- (c) With reference to an organisation that you have studied, explain how the Australian food industry contributes to the Australian economy.

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Name of organisation:

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End of Question 21

Please turn over

Question 22 (12 marks)

A gourmet food magazine is producing fresh food boxes which contain all of the ingredients to make their featured recipes. Customers can purchase the boxes as part of their magazine subscription.

- (a) Outline the reasons for the development of this food product.

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- (b) Explain quality control procedures that could be applied to guarantee freshness of the gourmet food box upon delivery to the consumer.

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Question 22 continues on page 13

Question 22 (continued)

- (c) Propose a suitable marketing plan to launch these gourmet food boxes.

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End of Question 22

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Question 23 (8 marks)

Refer to the following article to answer Question 23.

**The sharing economy comes
to a kitchen near you**

Food delivery company, *Yum to You*, is into the home meal delivery business. Instead of sourcing meals from traditional restaurants, *Yum to You* is tapping into the home-cooked market and sourcing meals directly from the kitchens of passionate home cooks who are looking for a way to increase their income while sharing their love of good food.

One example is Julie Jones who is famous among family and friends for her chicken pies. Julie, who only uses organic and ethically-sourced ingredients, has had to employ her neighbour to help her prepare enough pies to keep up with demand.

- (a) Which level of operation best describes Julie Jones's business?

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Question 23 continues on page 15

Do NOT write in this area.

Question 23 (continued)

- (b) Outline TWO legislative requirements that need to be met by Julie Jones's business. **3**

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- (c) Explain the storage and distribution systems that could be used by *Yum to You* to ensure food is delivered promptly from the home kitchen to the consumer while maintaining maximum quality. **4**

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End of Question 23

Question 24 (5 marks)

Discuss the relationship between the consumption of nutritionally modified foods and the health of the individual.

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Question 25 (8 marks)

- (a) Describe THREE functions of packaging.

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- (b) Explain how current developments in packaging can assist the elderly in meeting their food needs.

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Question 26 (8 marks)

How do both the Australian food industry and consumers benefit from the production of organic foods OR genetically-modified foods?

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Section III — Answer Question 27 on pages 19–25

27

Question Number

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If you require more space to answer Question 27, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 27, tick here.

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Section IV — Answer Question 28 on pages 26–32

28

Question Number

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If you require more space to answer Question 28, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 28, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 19–25 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

Farmers in southern New South Wales have a large quantity of hail-damaged apples that are unsuitable for sale in supermarkets due to their unsightly appearance.

A small, not-for-profit food processing company that is concerned about the environmental impact of food wastage, has offered to process the fruit to make it suitable for export to countries where the food supply is limited.

- | | | |
|-----|---|----------|
| (a) | Outline a food product that could be produced using the hail-damaged apples. | 2 |
| (b) | Describe TWO internal and TWO external factors that may affect the development and production of the product using the hail-damaged apples. | 5 |
| (c) | Explain a manufacturing process that would transform the hail-damaged apples into a product suitable for export as identified in part (a). | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 26–32 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Evaluate the role played by organisations that promote nutritional health in relation to both food consumption and diet-related conditions.

End of paper